

Probability refresher for ML

Why a model should say how sure it is, and the probability behind that, up to Bayes' rule.

THE QUESTION

A classifier meets an input far from its training data; a forecast runs past the last measurement. How does a model say how sure it is?

THE INTUITION

Treat every unknown, the parameters θ and the prediction y^* , as a random variable. The rules of probability then turn data into a posterior over θ , and predictions average over what is still unknown.

KEY TAKEAWAY

$p(\theta \mid D) = p(D \mid \theta)p(\theta)/p(D)$ runs through the whole course. MLE, MAP, conjugate priors, VI, Laplace and sampling each use or approximate it.

outcomes Ω → events $A \subseteq \Omega$ → $X: \Omega \rightarrow \mathbb{R}$ → PMF or PDF → joint → marginal, conditional → Bayes: $p(\theta \mid D)$

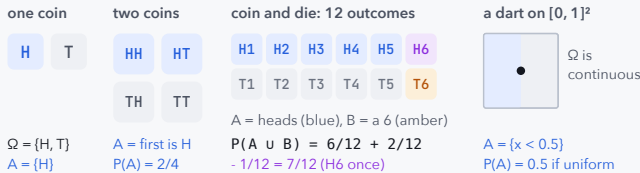
1 Uncertainty belongs in the prediction

illustrative (ours): a Bayesian straight-line fit to 10 points, known noise



The course sets out to predict with uncertainty, optimise black-box functions, build training sets efficiently and model how data is generated. Its opening examples: a classifier trained on two interleaved moons should output about 0.5 far from both classes, and a forecast of atmospheric CO₂ should widen with the horizon, as the band does here.

2 Outcomes, events and three axioms



The sample space Ω holds every outcome; an event is a subset $A \subseteq \Omega$. Axioms: $P(A) \geq 0$, $P(\Omega) = 1$, and $P(\bigcup_i A_i) = \sum_i P(A_i)$ for disjoint events. So $P(\emptyset) = 0$, $P(A^c) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. For three events, subtract all three pairwise overlaps and add back $P(A \cap B \cap C)$.

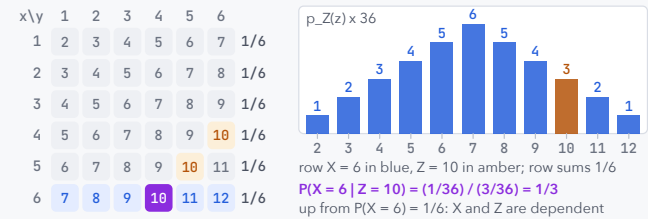
3 A random variable is a function on outcomes



$X: \Omega \rightarrow \mathbb{R}$; capital X is the variable, lowercase x a value it takes. One coin: $X(H) = 1$, $X(T) = 0$, or a gain $Y(H) = 1$, $Y(T) = -1$. A discrete X has a PMF $p_X(x) = P(X = x)$, such as the Bernoulli $p^x(1-p)^{1-x}$; a continuous one has a density, with $P(a \leq X \leq b) = \int_a^b p(x) dx$.

4 Two dice: joint, marginal and conditional

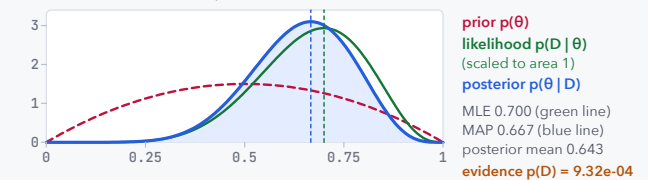
36 equally likely (x, y) ; cells show $z = x + y$



Each cell has joint probability $p(x, y) = 1/36$. A marginal sums out the other variable: $p_X(1) = \sum_y p(1, y) = 1/6$. The sum $Z = X + Y$ has $p_Z(z) = (6 - |z - 7|)/36$ for $z = 2, \dots, 12$. Conditioning keeps only the cells where B holds: $P(A \mid B) = P(A \cap B)/P(B)$. X and Y are independent, since $1/36 = 1/6 \cdot 1/6$; X and Z are not.

5 Bayes' rule, written for parameters

illustrative (ours): $\theta = P(\text{heads})$, prior Beta(2, 2), data 7 heads in 10 tosses



$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}, \quad p(D) = \int p(D \mid \theta)p(\theta) d\theta$$

Posterior equals likelihood times prior over evidence. On the dice, $P(X = 6 \mid Z = 10) = (1/6 \cdot 1/6)/(3/36) = 1/3$. Here a Beta prior and a Bernoulli likelihood give a Beta(9, 5) posterior: a conjugate pair, so $p(D) = 6B(9, 5) = 9.3 \times 10^{-4}$ in closed form.

6 One equation, many methods

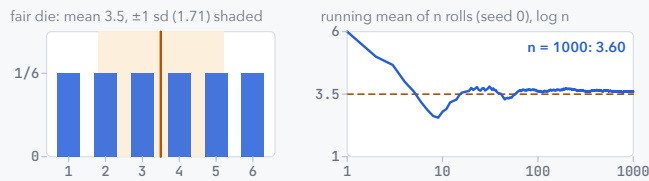
MLE	$\arg \max_{\theta} p(D \mid \theta)$: the likelihood alone
MAP	$\arg \max_{\theta} p(D \mid \theta)p(\theta)$: the prior joins in
Conjugate	the posterior stays in the prior's family
VI	$\arg \min_{\varphi} \text{KL}(q_{\varphi}(\theta) \parallel p(\theta \mid D))$
Laplace	$\mathcal{N}(\theta_{\text{MAP}}, \mathbf{H}^{-1})$, $\mathbf{H} = -\nabla^2 \log p(\theta \mid D)$
Sampling	draw $\theta_s \sim p(\theta \mid D)$ without $p(D)$

The obstacle is $p(D)$, an integral over every θ : intractable in most models. Predictions average over the posterior, and with samples $p(y^* \mid x^*, D) \approx \frac{1}{S} \sum_s p(y^* \mid x^*, \theta_s)$.

Moments, distributions, the normal and derivatives

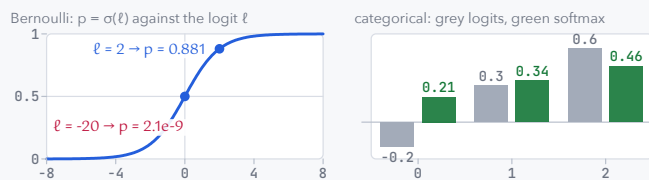
Moments, the notebooks' distributions, log space, the multivariate normal and the derivatives later sheets use.

7 Expectation, variance and covariance



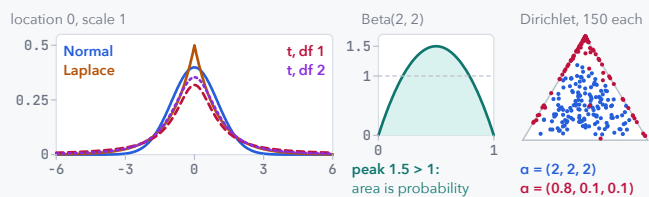
$E[X] = \sum_x x p_X(x)$ or $\int x p(x) dx$. Bernoulli: $E[X] = p$, $\text{Var}[X] = p(1-p)$. A fair die: $E[X] = 3.5$ and $\text{Var}[X] = E[X^2] - E[X]^2 = 91/6 - 12.25 = 35/12$. Expectation is linear; variances of independent variables add, so $\text{Var}[X+Y] = 35/6$. $\text{Cov}(X, Z) = E[XZ] - E[X]E[Z] = 35/12$ for $Z = X+Y$. np.std divides by n and pandas'.std() by $n-1$: 1.708 against 1.871 for the values 1 to 6.

8 Bernoulli and categorical, via logits



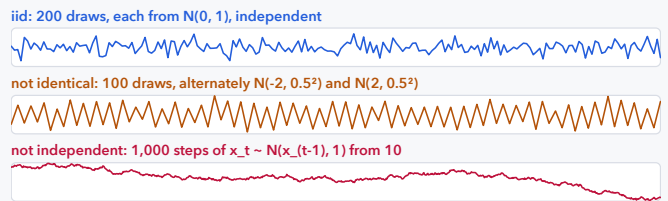
A probability lives in $[0, 1]$, a logit on the whole real line: $p = \sigma(\ell) = 1/(1 + e^{-\ell})$ and $\ell = \log p/(1-p)$. Libraries accept either, so check which you pass: logit -20 gives $p = 2.1 \times 10^{-9}$. A categorical over K classes takes K logits through a softmax, $p_k = e^{\ell_k} / \sum_j e^{\ell_j}$: $(-0.2, 0.3, 0.6)$ gives $(0.205, 0.338, 0.457)$. A logit can be negative and logits need not sum to 1, so plot the softmax output as the PMF, never the logits.

9 A density is not a probability



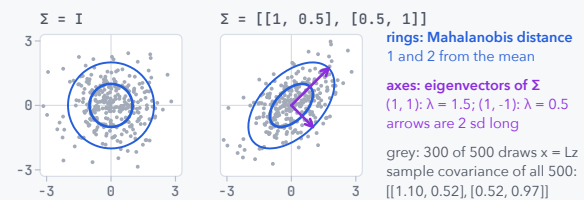
$p(x) \geq 0$ and $\int p(x) dx = 1$, but $P(X = x) = 0$ and $p(x)$ may exceed 1. Laplace and Student t have heavier tails than the normal: at $x = 5$, 1.5×10^{-6} against 0.0034 and 0.012 (t , df 1). Uniform(a, b) is $a + (b-a)U$; the notebook's generator $x \leftarrow (1664525x + 1013904223) \bmod 2^{32}$, seed 42, fills each of 10 bins with 939 to 1039 of 10,000 draws.

10 iid likelihoods multiply, so add logs



For iid data (identically distributed, independent), $p(D | \theta) = \prod_i p(x_i | \theta)$, so $\log p(D | \theta) = \sum_i \log p(x_i | \theta)$. Products underflow: for $\mathcal{N}(0, 1)$, $\log p(-20) = -200.9$, so p is 5.5×10^{-88} in float64 and 0 in float32 (the notebook, quoted). Over 1,000 seeded draws the product of densities is 0.0 and the sum of logs -1426.4 : use `log_prob`.

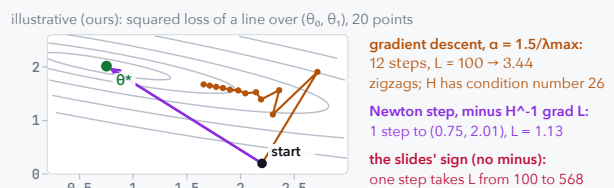
11 The univariate and multivariate normal



$$\mathcal{N}(x | \mu, \Sigma) = ((2\pi)^d |\Sigma|)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$

With $d = 1$: $(2\pi\sigma^2)^{-1/2} e^{-(x-\mu)^2/2\sigma^2}$. Σ must be symmetric positive definite; sample $x = \mu + Lz$ with $z \sim \mathcal{N}(0, I)$ and $LL^T = \Sigma$. The bivariate notebook's separate sliders for σ_{01} and σ_{10} allow unequal values: not a covariance.

12 Gradient, Jacobian and Hessian



derivative $3x^2 \rightarrow 6x$, which is 12 at $x = 2$
gradient $2x^2 + 3y \rightarrow (4x, 3) = (8, 3)$ at $(2, 1.5)$
Jacobian $(x^2 + y^2, y - z) \rightarrow [2x, 2y, 0; 0, 1, -1]$
Hessian $x^2 + y^2 + xyz \rightarrow [2, z, y; z, 2, x; y, x, 0]$

The Jacobian of $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is $m \times n$; the Hessian is the Jacobian of ∇f . For $L = \frac{1}{2} \sum_i r_i^2$, $r_i = y_i - \theta_0 - \theta_1 x_i$: $\nabla L = -(\sum r_i, \sum r_i x_i)$, $H = [n, \sum x_i; \sum x_i, \sum x_i^2]$. Newton's step $-H^{-1} \nabla L$ lands on the minimum of a quadratic. The slides drop the minus sign in ∇L .

$p(\theta | D) \propto p(D | \theta)p(\theta)$ • marginals sum, conditionals divide • add log probabilities • Σ symmetric positive definite

Next: the maximum likelihood estimation sheet. Source: lecture slides "Introduction", "Probability refresher" and "Calculus", and the distributions, bivariate normal and uniform random number notebooks (nipunbatra.github.io/pml-teaching). The line fit, coin posterior, loss surface and random draws are ours (NumPy, seeded); torch outputs are quoted; the generator's bin counts match the notebook. Corrected: the slide's sum for $p_X(1)$, the sign of the squared-loss gradient, and the notebook's plot of logits as probabilities.